

Cody Pinkston

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EDUCATION

Texas Christian University

Bachelor of Arts in Computer Science, Minor in Mathematics, John V. Roach Honors College
GPA: 3.7

Fort Worth, TX

Aug. 2023 – Present

TECHNICAL SKILLS

Languages: Java, Python, C/C++, MySQL, JavaScript, HTML/CSS

Frameworks: Node.js, Vue, Tailwind, Spring Boot

Developer Tools: Git, Docker, Maven, VS Code, IntelliJ, Power Apps, Power Automate

Libraries: CuPy, Thrust, Selenium

PROJECTS

GPU-Accelerated Ray Tracer | C++, CUDA, GLTF

Mar. 2026 – May 2026

- Collaborated to build a ray tracer rendering GLTF 3D models to images, supporting both a sequential CPU and a parallel GPU via CUDA
- Implemented Möller–Trumbore ray-triangle intersection, barycentric normal interpolation, and Lambertian + ambient shading with hard shadows
- Designed a configurable CLI supporting runtime device selection, custom camera/light positioning, and render resolution customization
- Built a benchmarking mode measuring per-phase (upload/kernel/download) GPU timing across multiple runs to quantify parallel speedup

C++ Shell | C++

Jan. 2026 – Feb. 2026

- Engineered a fully-functional Unix shell in C++ supporting interactive and batch execution modes.
- Built parallel command execution with the & operator
- Implemented I/O redirection (>) using open() + dup2() to redirect both stdout and stderr to a file.

TV Database | Python, mySQL

Oct. 2025 – Dec. 2025

- Designed and implemented a relational MySQL database of 10,000+ TV shows across 6 tables
- Applied database normalization (up to 3NF) to eliminate redundancy in genre and country data
- Bridged database to a Python GUI, supporting full CRUD functionality for intuitive data management

Movie Recommender | Java, Maven, Swing, Web Scraping

Dec. 2024 – Jan. 2025

- Developed a desktop application powered by user data from Letterboxd to algorithmically recommend movies
- Used JSoup to seamlessly pull user and film data directly from Letterboxd
- Built into a JAR using Maven and then an .exe installer for easy installation
- Crafted a multi-page Swing GUI incorporating filter selection, flexible viewing modes, and help page

EXPERIENCE

Application Developer

June 2026 – Aug. 2026

Texas Parks and Wildlife Department

Austin, TX

- Designed and developed a full-stack time-off request application using Power Apps, Power Automate, and SharePoint Lists, consolidating the request, review, and approval process into a single centralized tool
- Built automated approval workflows with Power Automate that route requests to managers and sync approved time off directly to a shared calendar, eliminating manual scheduling steps
- Authored a process flow document mapping the end-to-end application logic, and a UI wireframe document outlining the app's interface, ensuring clarity for stakeholders before development

LEADERSHIP AND VOLUNTEERING

Peer Mentor & Teaching Assistant, TCU College of Science & Engineering

Aug. 2026 – Present

Honors College Mentor

Aug. 2026 – Present

Furry Friendly Animal Rescue

June 2023 – Present

Honors College Community Service

Aug. 2025 – Present

Intramural Photographer

Aug. 2023 – May 2024